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A RAMCHARAN

GitHub
LinkedIn
Portfolio

Skills

Programming Languages: Python, SQL
Machine Learning & AI: Scikit-learn, Feature Engineering, Classification, Model Evaluation, TF-IDF, Logistic Regression
Deep Learning & Computer Vision: TensorFlow, Keras, PyTorch, CNN, MobileNetV2, Transfer Learning, Image Classification, Data Augmentation, Fine-Tuning, Grad-CAM
Generative AI: LangChain, RAG, LLMs, Prompt Engineering, Google Gemini API, OpenAI API, Hugging Face
Vector Search & Retrieval: ChromaDB, Vector Embeddings, Semantic Search, Contextual Compression, Cohere Reranking
Data Science: Pandas, NumPy, Matplotlib, Seaborn
Tools: Flask, REST APIs, Git, GitHub, Postman

Work Experience

Machine Learning Engineer Intern **Mar 2026 – May 2026**
Pratinik Infotech *India*

- Engineered features across **10+ industrial and transactional datasets**, cutting preprocessing time by **25%**.
- Built a **predictive maintenance model** using **Random Forest**, forecasting equipment failures with **82% accuracy**.
- Developed **fraud detection classifiers**, flagging anomalous transactions with **90% precision** across **100+ records**.

Projects

Chest X-Ray Pneumonia Classification **Aug 2026**
Python, TensorFlow, Keras, MobileNetV2, Scikit-learn, NumPy, Pandas, Grad-CAM

- Built a deep learning image classification pipeline using **ImageNet-pretrained MobileNetV2** to classify pediatric chest X-rays as NORMAL or PNEUMONIA.
- Applied **224x224 image preprocessing**, MobileNetV2 normalization, and training-only augmentation (rotation, translation, zoom).
- Achieved **78.04% test accuracy**, **84.93% pneumonia F1-score**, and **94.69% ROC-AUC** on a test set of **624 images**.
- Reached **98.97% pneumonia recall** and implemented **Grad-CAM** visualizations to interpret model predictions.

RAG Chatbot – Retrieval-Augmented Generation **Aug 2026**
Python, Streamlit, LangChain, ChromaDB, Cohere, OpenAI, Google Generative AI, Hugging Face

- Built a **Streamlit-based RAG chatbot** supporting PDF, TXT, DOCX, and CSV documents for context-grounded question answering.
- Implemented document chunking, vector embeddings, **ChromaDB** semantic retrieval, contextual compression, and **Cohere reranking**.
- Integrated **OpenAI, Google Generative AI, and Hugging Face** models to generate context-aware responses from retrieved documents.
- Achieved **82% answer correctness**, **84% faithfulness**, **81% answer relevancy**, **79% context precision**, and **83% context recall**.

Medical Chatbot – AI Symptom Triage Assistant **Jun 2026 – Jul 2026**
Python, Flask, Scikit-learn, TF-IDF, Logistic Regression, Gemini API, Pandas, Joblib, Gunicorn

- Built a **Flask-based AI symptom triage app** combining an ML classifier with **Google's Gemini API** for plain-language health guidance.
- Developed a **TF-IDF + Logistic Regression** classification pipeline using Scikit-learn for symptom-based prediction.
- Integrated **Gemini API** for contextual explanations, keeping ML predictions separate from generative responses.

Education

Lovely Professional University **Aug 2024 – Jun 2028**
Bachelor of Technology in Computer Science and Engineering — 9.22 CGPA *Jalandhar, Punjab*
Specialization: Generative AI

Sai Sri Chaitanya Junior College **2022 – 2024**
Mathematics, Physics & Chemistry — 95.3% *Andhra Pradesh, India*

Certifications

- **Azure AI Engineer Associate (AI-102)** — Microsoft
- **Azure AI Fundamentals (AI-900)** — Microsoft
- **Azure Data Fundamentals (DP-900)** — Microsoft
- **Azure Fundamentals (AZ-900)** — Microsoft

Achievements

- Led teams in **3+ hackathons**, qualified for **2 rounds**, and participated in national-level hackathons at **IIT Jodhpur** and **IIT Ropar**.